PV Mounting Structure – Request for Quotation

Doc. PV-RFQ-STR-001 Rev A
3 November 2026

185.4 kWp rooftop PV · Marriott Hotel, mezzanine roof · Uganda

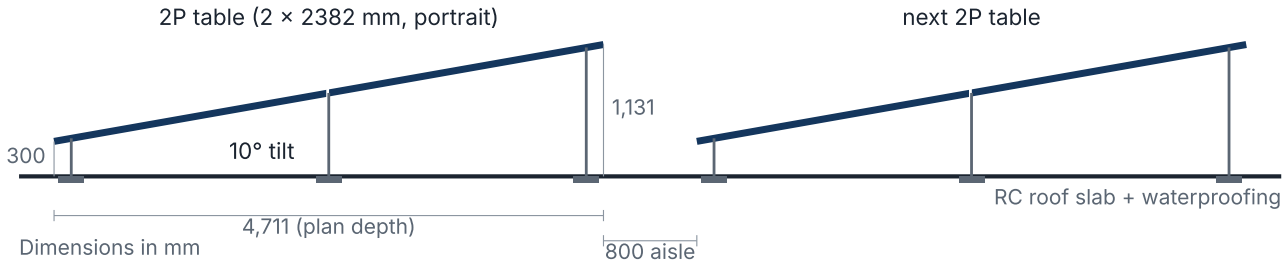
Please quote for the design and supply of the PV mounting structure below, including the structural calculation.

1. System and module

Solar capacity	**185.4 kWp**
Number of modules	**299 modules × 620 Wp**
Module size / weight	**2382 × 1134 × 30 mm**, ≈33 kg each (bifacial, dual glass)
Total module weight	≈9.9 t (≈12.2 kg/m² of module area)

2. Structure configuration

Tilt	**10°** fixed
Module orientation	Portrait, 2 modules high per table (2P)
Number of tables	**6 tables of 2P × 23 modules (46 each) + 1 table of 1P × 23 modules**
Table length	26,522 mm
Table depth on plan	4,711 mm (2P) / 2,346 mm (1P)
Front / rear edge height	300 mm / ≈1,131 mm above roof (2P)
Spacing	20 mm between modules · 800 mm aisle between tables · 1,000 mm from parapet



Typical section through two 2P tables

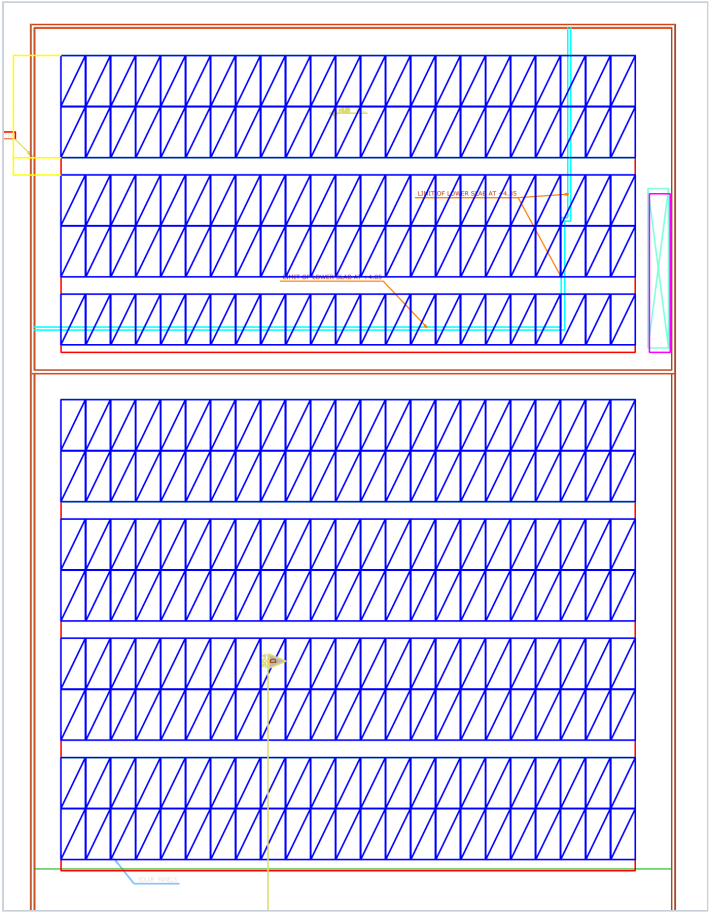
3. Wind load and structural design

Basic wind speed	**25 m/s** (10-min mean at 10 m; preliminary, supplier to confirm for the site)
Peak wind pressure	**q_p ≈ 0.67 kN/m²** (EN 1991-1-4, terrain cat. III, roof height ≈7–10 m)
Site exposure	Two hotel towers stand next to the roof: allow for wind acceleration and edge/corner zones at the parapet
Design codes	EN 1991-1-4, EN 1999 (aluminium) or ASCE 7 equivalent; design life 25 years
Materials	Aluminium 6005-T6 / 6063-T6; stainless-steel fasteners A2-70; corrosion category C3
Module clamping	To the module installation manual (rated test loads 5,400 Pa down / 2,400 Pa up)

4. Roof and fixing

Roof	Flat reinforced-concrete slab, single level, waterproofed, with parapet. Usable area ≈29.4 × 39 m.
Fixing (please quote both)	**A – Anchored (preferred):** chemical anchors or concrete plinths, with waterproofing detail **B – Ballasted:** concrete ballast with wind deflectors; state ballast weight per table
Roof protection	EPDM pads under every foot; keep drainage paths clear
Roof load	State the uniform load (kg/m²) and the point load per foot (kN), for approval by the building's structural engineer

5. Layout



PV layout on the mezzanine roof (DWG/DXF attached)

6. Quotation to include

1. Price (USD per Wp and total), delivery terms and lead time
2. Structural calculation: wind loads, loads per foot, anchor or ballast design
3. General arrangement drawings and bill of materials
4. Warranty (structure 10 years minimum) and installation manual

Contact: [Name / Company / Phone / Email]